# An all-atom structure aware method for versatile prediction of protein stability changes and broad mutational effects

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## ABSTRACT

Accurate prediction of protein stability changes upon mutations is crucial for biomedical research and protein engineering. Existing methods for protein stability prediction often rely on backbone atomic coordinates, neglecting the significant structural changes in protein side chains induced by mutations. Here, we present FA-stab, an full-atom-aware model for predicting changes in protein stability ($\Delta\Delta$G). FA-stab employs Euclidean Neural Networks (e3nn) to capture the three-dimensional positions of all atoms, including side-chain atoms, and further augments its predictions with evolutionary information derived from protein language models. This approach achieves high prediction accuracy, attaining state-of-the-art correlation coefficients on the S669 benchmark dataset and outperforming backbone-based models such as ThermoMPNN. Furthermore, on a large-scale multi-mutation dataset of the orphan protein SeviL, FA-stab demonstrates the advantage of leveraging full structural information, surpassing models that rely on backbone structure information (SPIRED-Stab) or statistical potentials (DDGun3D). Finally, we extend the FA-stab architecture to predict mutational effects on a broader scale, developing FA-fitness applied to deep mutational scanning (DMS) data. FA-fitness shows promising performance in predicting mutation pathogenicity. Thus, this study presents a versatile, full-atom aware framework for predicting mutational effects in proteins.


***Keywords*** protein stability · protein fitness · mutation effect

## Introduction

Proteins perform a wide array of essential functions within living organisms, including the replication and expression of genetic information, catalysis of biochemical reactions, transport of substances, and cellular signal transduction. Protein stability is crucial for proper folding into functional three-dimensional structures and for maintaining normal biological activity. Understanding the impact of mutations on protein stability holds significant importance for protein engineering, protein design, and fundamental biomedical research. For instance, in protein engineering, directed evolution is often employed to develop enzymes that remain stable and functional at elevated temperatures. Meanwhile, identifying how mutations affect the stability of human proteins is vital for elucidating mutational pathogenicity and disease mechanisms. Wet-lab experiments for large-scale determination of mutational effects on protein stability are both time-consuming and costly, given the vastness of protein sequence space. Therefore, developing accurate predictive models for protein stability is of great significance for fundamental life science research and biotechnological applications[1].

Numerous models have been developed in previous research to predict changes in protein stability upon mutation. The change in Gibbs free energy ($\Delta\Delta$G) serves as a key metric for quantifying protein stability changes, representing the difference in Gibbs free energy ($\Delta$G) between the mutant and wild-type proteins. Researchers have implemented $\Delta\Delta$G prediction using statistical potentials as well as machine learning approaches, including FoldX[2], DDGun[3] and DUET[4]. With the rise and advancement of deep learning, a growing number of protein stability prediction models based on deep learning have emerged. These models primarily rely on either sequence information or protein structural information to predict mutational effects. For sequence-based models, inputs typically include evolutionary information extracted by protein language models or derived from multiple sequence alignments (MSA). For instance, PROSTATA[5] and Mutate Everything[6] utilize Evolutionary Scale Modeling (ESM)[7] as an evolutionary feature extractor. Structure-based models, on the other hand, employ structural encoders to extract protein structural features as critical input for inference. For example, Pythia[8] uses a message-passing neural network (MPNN) to encode protein backbone geometry, while ThermoMPNN[9] applies transfer learning by adapting the pre-trained protein inverse folding model ProteinMPNN[10] to the task of stability prediction, achieving high $\Delta\Delta$G prediction accuracy. Additionally, hybrid models that integrate both structural and sequence information have also been developed. Examples include GeoDDG[11] and SPIRED-Stab[12], which employ graph neural networks to jointly process protein structural and sequence features.

However, existing protein stability prediction models often rely solely on backbone information, neglecting the atomic details of side chains. Mutant protein structures largely retain the wild-type backbone conformation. The most significant structural differences between wild-type and mutant proteins often arise from the expansion or contraction of side-chain conformation, a critical source of variation that remains uncaptured by the aforementioned models. The spatial features of protein side-chain atoms significantly influence protein-protein interactions, protein-nucleic acid binding, and the affinity of proteins for small-molecule drugs. Recent algorithmic advances, such as the Euclidean Neural Networks (e3nn)[13] framework, have enabled the development of equivariant networks capable of all-atom encoding of biomolecules like proteins and nucleic acids.

To fully exploit the information from side-chain changes before and after mutation, we developed an all-atom aware protein stability prediction model named FA-stab. This model utilizes the e3nn network to extract comprehensive atomic structural information from both wild-type and mutational states and integrates it with evolutionary information provided by protein language models, achieving high accuracy in $\Delta\Delta$G prediction. Furthermore, we expanded this all-atom aware framework to protein fitness prediction, resulting in FA-fitness. The models, FA-fitness and FA-stab, demonstrated strong predictive performance across different mutation effect prediction tasks, including stability changes, fitness and pathogenicity.

## Results

### The Model Architecture of the FA-stab and FA-fitness

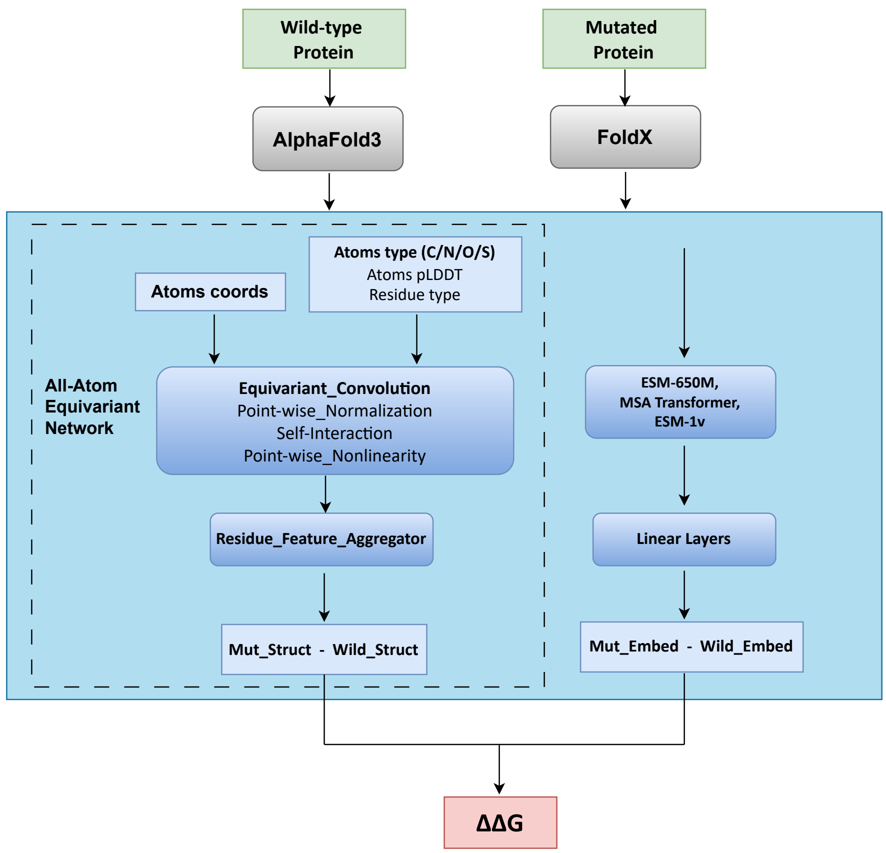


**Figure 1. Schematic of the FA-stab model architecture.** The network enclosed by the dashed box represents the all-atom equivariant network, which serves as the core component of FA-stab. It encodes the wild-type protein structure predicted AlphaFold3 and the mutant structure constructed by FoldX at the all-atom level, producing structural feature representations. Sequence features for both wild-type and mutant proteins are extracted by protein language models and subsequently processed through shallow linear layers. The differences in structural and sequence features (mutant minus wild-type) are calculated separately, then combined. The feature differences at the mutated site are extracted and summed to obtain the final $\Delta\Delta G$.

FA-stab accepts two all-atom protein structures as input: the wild-type structure predicted by AlphaFold3[14] and the mutant structure modeled by FoldX[2]. The core component of FA-stab is a protein structure encoder named all-atom equivariant network, which extracts spatial features of protein structures at the all-atom level. This structure encoder consists of three blocks, where each block contains an equivariant convolutional layer, Point-wise Normalization, Self-Interaction, and Point-wise Nonlinearity. The equivariant convolution extracts inter-atomic distance and orientations information from the input atomic coordinates, subsequently mixing and interacting this spatial positional information with atomic type features. The atom-level feature representations processed by the all-atom equivariant network are then aggregated into residue-level structural features. Additionally, FA-stab encodes sequence embedding obtained from language models (ESM-650M[7], MSAtransformer[15], ESM1v[16]) through shallow linear layers to derive evolutionary information features of the protein sequence. The feature differences are calculated by subtracting the wild-type protein's structural/sequence features from those of the mutant variant. These differences are then combined using learnable parameters to generate the final feature representation for each residue site. The final features corresponding to the mutation sites are extracted and summed to produce the final predicted $\Delta\Delta$G value.

The network architecture of FA-fitness is similar to that of FA-stab, as illustrated in Supplementary Figure S1. However, FA-fitness incorporates a Graph Attention Network which uses residue-level structural features as node attributes and the contact probability along with the predicted aligned error (pAE) from AlphaFold3 as edge attributes, enabling further integration and refinement of structural information. Another key distinction is that FA-fitness takes only the all-atom structure of the wild-type protein as its input and outputs predicted fitness scores for all possible single-point mutations.

### FA-stab performs well on $\Delta\Delta$G test sets S669

To validate the performance of FA-stab, we compared it with other prediction models on the widely adopted $\Delta\Delta$G benchmark dataset S669[17]. The protein sequences in the S669 test set share less than 30% similarity with those in the training set of FA-stab, ensuring an unbiased evaluation. The Spearman's correlation coefficient (SCC), Pearson's correlation coefficient (PCC), Kendall's correlation coefficient (KCC), root mean square error (RMSE) and mean absolute error (MAE) were used to evaluate both the correlation and the numerical error between predicted $\Delta\Delta$G and experimentally measured $\Delta\Delta$G. In most previous studies, correlation coefficients have typically been computed using all stability data of mutations across the entire test set. However, in practical protein engineering, researchers are often more concerned with the correlation of $\Delta\Delta$G predictions for mutations within the same protein. Therefore, in this study, we calculated SCC, PCC, and KCC specifically for mutations grouped by individual protein. To ensure the reliability of the correlation estimates, proteins with fewer than 10 mutations in the S669 test set were excluded. After computing SCC, PCC, and RMSE for each qualifying protein, the mean and standard deviation of these metrics across all retained proteins were then determined.

As shown in Table 1, we compared 17 structure-based methods, 8 sequence-based methods, and 5 hybrid approaches that integrate both structural and sequence information. Among all the models, FA-stab achieves the highest SCC (0.62) and PCC (0.62), outperforming ThermoMPNN (SCC=0.56 and PCC=0.53) which uses protein backbone structure as input. FA-stab is the only model to exceed 0.6 in both SCC and PCC metrics. Additionally, the best performance in terms of KCC is attained by FA-stab and SPIRED-Stab, both reaching 0.46. Therefore, these results indicate that FA-stab exhibits strong correlation between predicted $\Delta\Delta$G and experimentally determined $\Delta\Delta$G. Furthermore, among the correlation metrics, the top-performing structure-based models are ThermoMPNN and Pythia, the best sequence-based models are PROSTATA and Mutate Everything, and the leading hybrid models are FA-stab and SPIRED-Stab. These models represent the state-of-the-art predictive performance across different categories of $\Delta\Delta$G prediction approaches. In terms of RMSE, FA-stab achieves 1.06 kcal/mol, which is close to the best-performing model in this metric, PROSTATA (1.05 kcal/mol). For MAE, the best results are obtained by PROSTATA and SPIRED-Stab, both reaching 0.82 kcal/mol, while FA-stab matched the accuracy of ACDC-NN, which has the lowest MAE among structure-based methods (0.87 kcal/mol).

In theory, $\Delta\Delta$G should satisfy antisymmetry, meaning that the $\Delta\Delta$G values for a forward mutation and its corresponding inverse mutation are opposite in sign. Suppose the change in protein stability upon direct mutation from sequence A to sequence B is denoted as $\Delta\Delta G_{\mathrm{AB}}$, and the stability change upon inverse mutation from sequence B back to A is denoted as $\Delta\Delta G_{\mathrm{BA}}$. Theoretically, it should hold that $\Delta\Delta \mathrm{G}_{\mathrm{AB}} = -\Delta\Delta \mathrm{G}_{\mathrm{BA}}$. Because some models fail to preserve this antisymmetric property in their predicted $\Delta\Delta \mathrm{G}$ , their accuracy significantly decreases when predicting $\Delta\Delta \mathrm{G}$ for inverse mutations[18]. Therefore, in this study, we evaluated the performance of FA-stab on inverse mutations. As shown in Table S1, models such as I-Mutant3.0, DUET, and SDM exhibit a significant decline in predictive accuracy for inverse mutations compared to their performance on direct mutations. Although FA-stab shows a minor decline in its metrics for inverse mutations, it nevertheless surpasses other models in terms of SCC (0.59), PCC (0.58), and RMSE (1.04 kcal/mol).

**Table 1. $\Delta\Delta \mathrm{G}$ prediction performace evaluated within each protein on the S669 test set**

| **Method** | **SCC** ↑ | **PCC** ↑ | **KCC** ↑ | **RMSE** ↓ | **MAE** ↓ |
|---|---|---|---|---|---|
| *Sequence-based* | | | | | |
| **MUPro**[19] | 0.18 ± 0.33 | 0.21 ± 0.34 | 0.13 ± 0.23 | 1.27 ± 0.44 | 1.06 ± 0.39 |
| **I-Mutant3.0-Seq**[20] | 0.22 ± 0.37 | 0.28 ± 0.33 | 0.16 ± 0.27 | 1.24 ± 0.41 | 1.03 ± 0.38 |
| **SAAFEC-SEQ**[21] | 0.34 ± 0.30 | 0.39 ± 0.29 | 0.26 ± 0.22 | 1.19 ± 0.46 | 0.97 ± 0.41 |
| **ACDC-NN-Seq**[22] | 0.39 ± 0.29 | 0.44 ± 0.29 | 0.29 ± 0.21 | 1.14 ± 0.47 | 0.90 ± 0.43 |
| **INPS-Seq**[23] | 0.40 ± 0.28 | 0.44 ± 0.26 | 0.28 ± 0.22 | 1.14 ± 0.50 | 0.92 ± 0.44 |
| **DDGun**[3] | 0.45 ± 0.27 | 0.47 ± 0.27 | 0.34 ± 0.21 | 1.39 ± 0.68 | 1.13 ± 0.60 |
| **Mutate Everything (ESM-2)**[6] | <u>0.50 ± 0.34</u> | 0.52 ± 0.30 | <u>0.38 ± 0.25</u> | 1.11 ± 0.52 | 0.89 ± 0.45 |
| **PROSTATA**[5] | <u>0.50 ± 0.28</u> | <u>0.53 ± 0.27</u> | 0.36 ± 0.22 | <u>**1.05** ± 0.50</u> | <u>**0.82** ± 0.43</u> |
| *Structure-based* | | | | | |
| **SDM**[24] | 0.31 ± 0.33 | 0.33 ± 0.36 | 0.22 ± 0.26 | 1.37 ± 0.34 | 1.08 ± 0.33 |
| **ThermoNet**[25] | 0.31 ± 0.30 | 0.38 ± 0.30 | 0.22 ± 0.23 | 1.21 ± 0.42 | 0.97 ± 0.40 |
| **mCSM**[26] | 0.33 ± 0.30 | 0.35 ± 0.31 | 0.23 ± 0.23 | 1.19 ± 0.49 | 0.97 ± 0.43 |
| **Dynamut**[27] | 0.34 ± 0.31 | 0.34 ± 0.32 | 0.25 ± 0.23 | 1.22 ± 0.46 | 0.97 ± 0.42 |
| **I-Mutant3.0**[20] | 0.34 ± 0.30 | 0.39 ± 0.29 | 0.26 ± 0.22 | 1.19 ± 0.46 | 0.97 ± 0.41 |
| **MAESTRO**[28] | 0.35 ± 0.33 | 0.40 ± 0.33 | 0.26 ± 0.25 | 1.14 ± 0.45 | 0.90 ± 0.38 |
| **DUET**[4] | 0.38 ± 0.30 | 0.41 ± 0.30 | 0.27 ± 0.24 | 1.16 ± 0.46 | 0.93 ± 0.41 |
| **ACDC-NN**[22] | 0.40 ± 0.30 | 0.42 ± 0.33 | 0.30 ± 0.23 | 1.10 ± 0.47 | <u>0.87 ± 0.43</u> |
| **FoldX**[2] | 0.41 ± 0.35 | 0.36 ± 0.39 | 0.31 ± 0.28 | 2.02 ± 1.19 | 1.52 ± 0.96 |
| **PoPMuSiC**[29] | 0.42 ± 0.29 | 0.45 ± 0.30 | 0.30 ± 0.23 | 1.15 ± 0.56 | 0.93 ± 0.49 |
| **DDGun3D**[3] | 0.43 ± 0.31 | 0.48 ± 0.29 | 0.32 ± 0.24 | 1.21 ± 0.59 | 0.93 ± 0.52 |
| **PremPS**[30] | 0.45 ± 0.29 | 0.48 ± 0.28 | 0.33 ± 0.24 | 1.18 ± 0.53 | 0.96 ± 0.46 |
| **DDMut**[31] | 0.46 ± 0.37 | 0.52 ± 0.28 | 0.35 ± 0.29 | <u>1.08 ± 0.50</u> | 0.88 ± 0.46 |
| **INPS3D**[23] | 0.46 ± 0.27 | 0.48 ± 0.28 | 0.35 ± 0.21 | 1.13 ± 0.52 | 0.90 ± 0.44 |
| **RaSP**[32] | 0.52 ± 0.29 | 0.50 ± 0.28 | 0.39 ± 0.23 | 1.24 ± 0.64 | 0.98 ± 0.59 |
| **Pythia**[8] | 0.55 ± 0.30 | <u>0.55 ± 0.30</u> | 0.41 ± 0.22 | 7.81 ± 2.94 | 6.34 ± 2.70 |
| **ThermoMPNN**[9] | <u>0.56 ± 0.31</u> | 0.53 ± 0.31 | <u>0.42 ± 0.25</u> | 1.15 ± 0.57 | 0.93 ± 0.51 |
| *Structure-sequence-hybrid* | | | | | |
| **GeoDDG-3D**[11] | 0.50 ± 0.28 | 0.51 ± 0.27 | 0.38 ± 0.22 | 1.11 ± 0.41 | 0.87 ± 0.39 |
| **GeoDDG-Seq**[11] | 0.51 ± 0.28 | 0.52 ± 0.29 | 0.39 ± 0.22 | 1.08 ± 0.42 | 0.85 ± 0.37 |
| **SPIRED-Fitness (zero-shot)**[12] | 0.54 ± 0.25 | 0.52 ± 0.27 | 0.41 ± 0.20 | 1.74 ± 0.55 | 1.50 ± 0.49 |
| **GeoStab v2**[12] | 0.55 ± 0.41 | 0.54 ± 0.41 | 0.43 ± 0.32 | <u>1.06 ± 0.51</u> | 0.84 ± 0.46 |
| **SPIRED-Stab**[12] | 0.58 ± 0.34 | 0.55 ± 0.36 | <u>**0.46** ± 0.27</u> | 1.07 ± 0.51 | <u>**0.82** ± 0.45</u> |
| **FA-Stab** | <u>**0.62** ± 0.17</u> | <u>**0.62** ± 0.14</u> | <u>**0.46** ± 0.14</u> | <u>1.06 ± 0.45</u> | 0.87 ± 0.42 |

All metrics are evaluated on a per-protein basis, with the final result presented as the mean ± standard deviation across all proteins. For each metric, the top-performing model is identified by comparing these mean values and is indicated in bold, while the best metric within each model category is marked with an underline.

It is noteworthy that both ThermoMPNN and Pythia have been pre-trained on large-scale protein structural datasets (e.g., the PDB[33] and CATH[34]), whereas FA-stab learned structural information from only 334 proteins of its training set. Despite this, FA-stab achieved higher prediction accuracy than these models on the S669 test set.

#### Performance evaluation on a large-scale multi-mutation dataset of 'orphan' protein

Evolution has produced numerous proteins with similar structures yet divergent sequences. These sequence variations encode vital evolutionary and co-evolutionary information. Models like AlphaFold achieve high-accuracy structure prediction precisely by harnessing the co-evolutionary signals embedded within multiple sequence alignments (MSA). However, for proteins lacking homologous sequences, such as orphan proteins, the prediction accuracy of AlphaFold declines. To address this, Tang et al. employed protein stability as a constraint to screen randomized mutants, generating a large collection of artificial homologous sequences (aMSA) with identical structures through an experimental approach termed Sibs-Seq[35]. Using the aMSA derived from Sibs-Seq, Tang et al. achieved higher AlphaFold2 prediction accuracy, including the orphan protein SeviL (PDB ID: 6LF2) with 129 residues.

In this study, we utilized the SeviL stability dataset from Sibs-Seq as a test set to evaluate the stability prediction performance on orphan proteins. The Sibs-Seq assay provides a fitness score for each mutation, which highly correlates with protein stability. In this scoring system, the wild-type protein is assigned a fitness score of 0. A mutant with a score less than 0 is considered less stable than the wild-type, whereas a score greater than 0 indicates a stabilizing mutation. The SeviL test set contains a large volume of multi-mutation data, including 891

As shown in Figure 2, FA-stab outperforms the SPIRED-Stab and DDGun models across all mutation types, single, double, triple, and higher-order mutations, in both SCC and PCC metrics. As shown in Supplementary Figure S2, S3 and S4, the kernal density plots comparing the predicted $\Delta\Delta$G against the experimental fitness scores from Sibs-Seq reveal that data points for FA-stab are more tightly clustered around the regression line, whereas those for the other three models show greater dispersion. This further indicates a stronger correlation between FA-stab predictions and experimental results. Additionally, the sequence-based DDGun performed poorly on the SeviL test set, with SCC and PCC values close to zero across all mutation categories. A key reason for this, as noted earlier, is DDGun's strong reliance on evolutionary information from MSA. In contrast, DDGun3D, which incorporates structural information, showed a substantial improvement over DDGun, underscoring the importance of structural features in maintaining or enhancing prediction accuracy for proteins with limited homologous sequences. Furthermore, FA-stab, which leverages all-atom structural features, achieved higher scores across all metrics compared to SPIRED-Stab, which uses only protein backbone coordinates. Additionally, as the mutational order increases from single to multiple mutations, the SCC and PCC values of FA-stab, SPIRED-Stab, and DDGun3D all decline, indicating that predicting $\Delta\Delta$G for multiple mutations remains a challenging task for models of different types. Finally, we evaluated the performance of the models on mutation data from all 8 proteins in the Sibs-Seq dataset (including 7 'non-orphan' proteins), as shown in Supplementary Tables S2, S3, and S4. Across all mutation types, FA-stab achieved higher average SCC over all proteins than the other models.

#### Evaluating FA-stab on stability benchmark datasets of disease-associated proteins

Myoglobin, an essential oxygen-storage protein in muscle tissue[36], and certain mutations in myoglobin can lead to muscle weakness and atrophy (Myoglobinopathy)[37]. Additionally, Myoglobin is commonly used in protein stability benchmarking[38]. After removing duplicate mutations from the original dataset, 113 mutations were retained for evaluation. On Myoglobin test set, FA-stab outperforms ThermoMPNN that relies on protein backbone coordinates

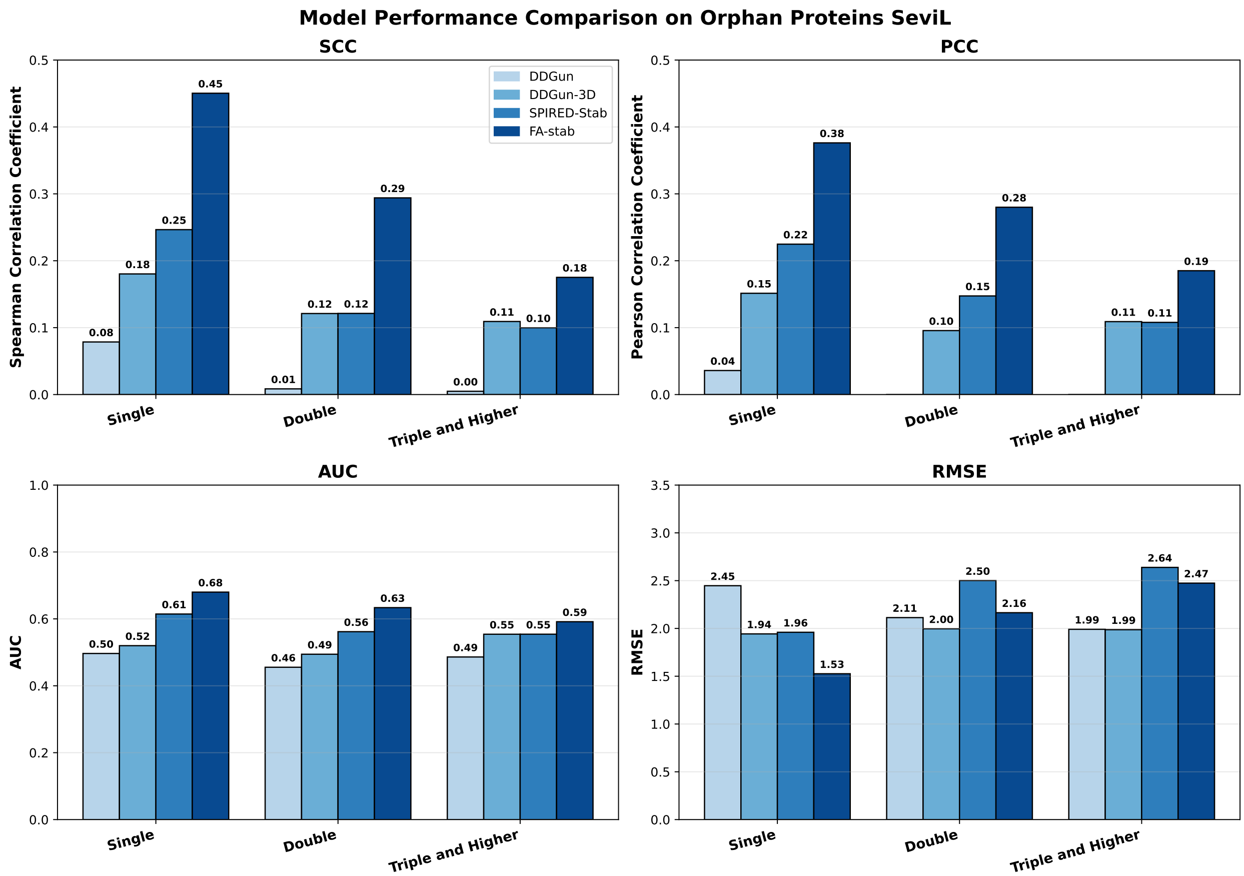


**Figure 2. Model performance on the orphan protein SeviL (PDB: 6LF2) test set.** Comparison of FA-stab, DDGun, DDGun3D, and SPIRED-Stab using PCC, SCC, AUC, and RMSE metrics across single mutations (N=891), double mutations (N=20332), triple and higher-order mutations (N=37911).

as input, achieving an SCC of 0.63, a PCC of 0.66, and an RMSE of 0.87 kcal/mol. In comparison, ThermoMPNN attains SCC and PCC values of only 0.53 and 0.52, respectively.

The tumor suppressor p53 plays a critical role in processes such as cell-cycle arrest and apoptosis[39]. Mutations in p53 contribute significantly to the development of various cancers, making the study of its mutational effects highly relevant for cancer therapeutics. Additionally, p53 serves as an important benchmark dataset for protein stability prediction[9]. In our evaluation on the p53 test set, FA-stab exhibits prediction performance comparable to that of ThermoMPNN. While FA-stab shows a slightly lower PCC than ThermoMPNN (0.70 vs. 0.72), it achieved a higher SCC (0.68 vs. 0.62).

Alpha-1-antitrypsin (AAT) is a serine protease inhibitor whose primary target is elastase. Alpha-1-antitrypsin deficiency (A1ATD) is one of the most common inherited disorders among individuals of European descent, with approximately 1 in 25 persons of European descent carrying at least one abnormal copy of the AAT. If a gene mutation leads to low AAT levels or misfolding of the protein, elastase can excessively degrade elastin in the pulmonary alveoli, thereby contributing to the development of emphysema[40]. On the Alpha-1-antitrypsin test set, the prediction accuracy of FA-stab is substantially higher than that of ThermoMPNN. As shown in the figure, FA-stab achieves an SCC of 0.46 and an RMSE of 1.22 kcal/mol, whereas ThermoMPNN attains an SCC of only –0.07 and an RMSE of 1.54 kcal/mol.

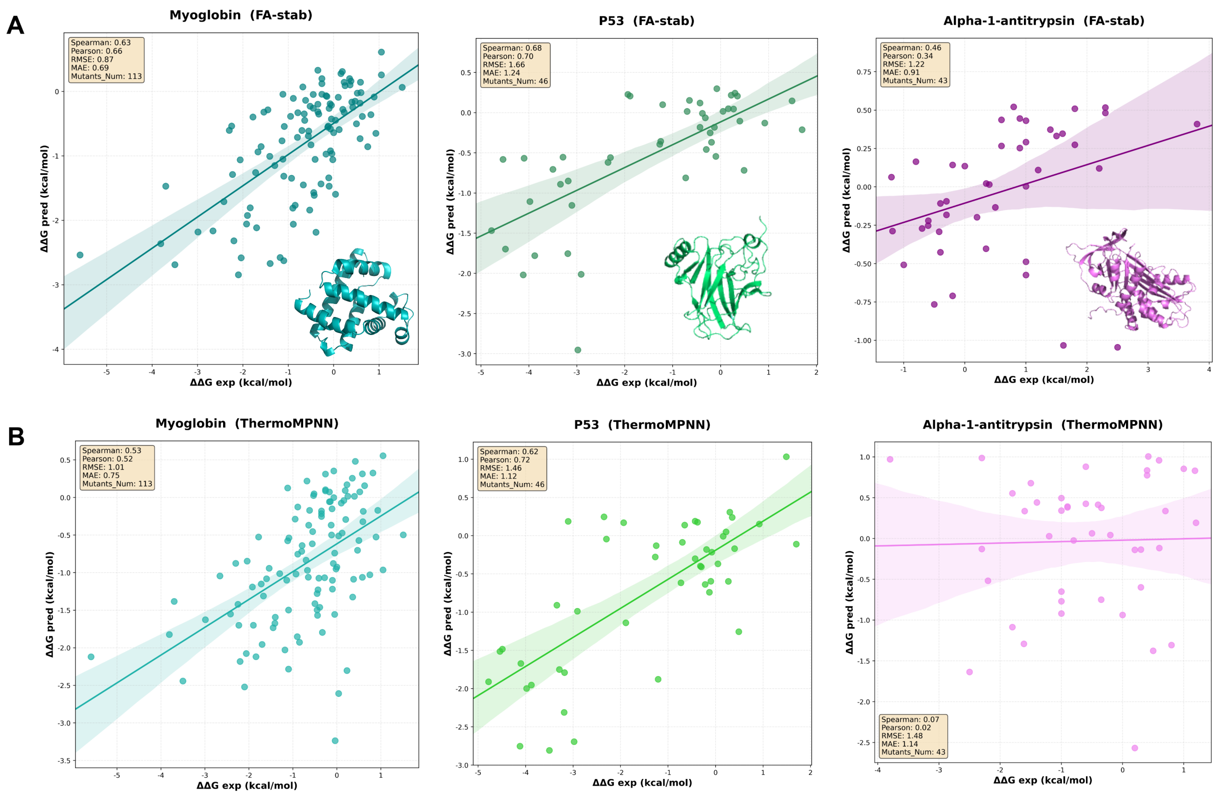


**Figure 3. Performance of FA-stab and ThermoMPNN on disease-associated protein stability test sets.** (A and B) Dot plots displaying the $\Delta\Delta G$ prediction results of FA-stab and ThermoMPNN for Myoglobin, p53, and Alpha-1-antitrypsin. The x-axis represents the experimentally determined $\Delta\Delta G$ (kcal/mol), and the y-axis represents the predicted $\Delta\Delta G$ (kcal/mol). The SCC, PCC, RMSE, and MAE values for each model on the corresponding test set are annotated in the legend.

### Evaluation of FA-fitness on fitness prediction and mutational pathogenicity prediction

Since mutations affect not only stability but also diverse protein properties, including enzymatic activity, expression levels, and subcellular localization. Thus, pathogenicity prediction approach based solely on stability will overlook other key mutatonal effects and show relative prediction accuracy. To overcome this limitation, we extended the FA-stab framework by training it on deep mutational scanning (DMS) data, yielding a model termed FA-fitness. The model architecture of FA-fitness is illustrated in Supplementary Figure S1, FA-fitness differs from FA-stab by accepting only the wild-type protein structure and sequence embeddings as input, enabling the prediction of fitness scores for every possible single-point mutation across the protein sequence. The training set integrates DMS profiles from 62 proteins (collected from MaveDB[41] and the DeepSequence Dataset[42]) with the MegaScale $\Delta\Delta G$ dataset.

To evaluate the predictive capability and generalization of FA-fitness on functional fitness scores, we constructed a zero-shot test set using proteins from the ProteinGym 'DMS Substitutions' benchmark that share less than 30% sequence similarity with the training data. As shown in Supplementary Table S5, FA-fitness achieves an average SCC of 0.48 across all proteins in this test set. Among the numerous zero-shot models reported in ProteinGym, FA-fitness ranks third, tying with S2F_MSA[43] (0.48) and ProSST-2048[44] (0.48), and closely approaches the top performers VenusREM[45] (0.49) and S3F_MSA[43] (0.49). FA-fitness outperforms models such as TranceptEVE_L[46] (0.46) and GEMME[47] (0.45). These results demonstrate that FA-fitness, trained on a limited set of proteins with DMS data, generalizes well to unseen proteins.

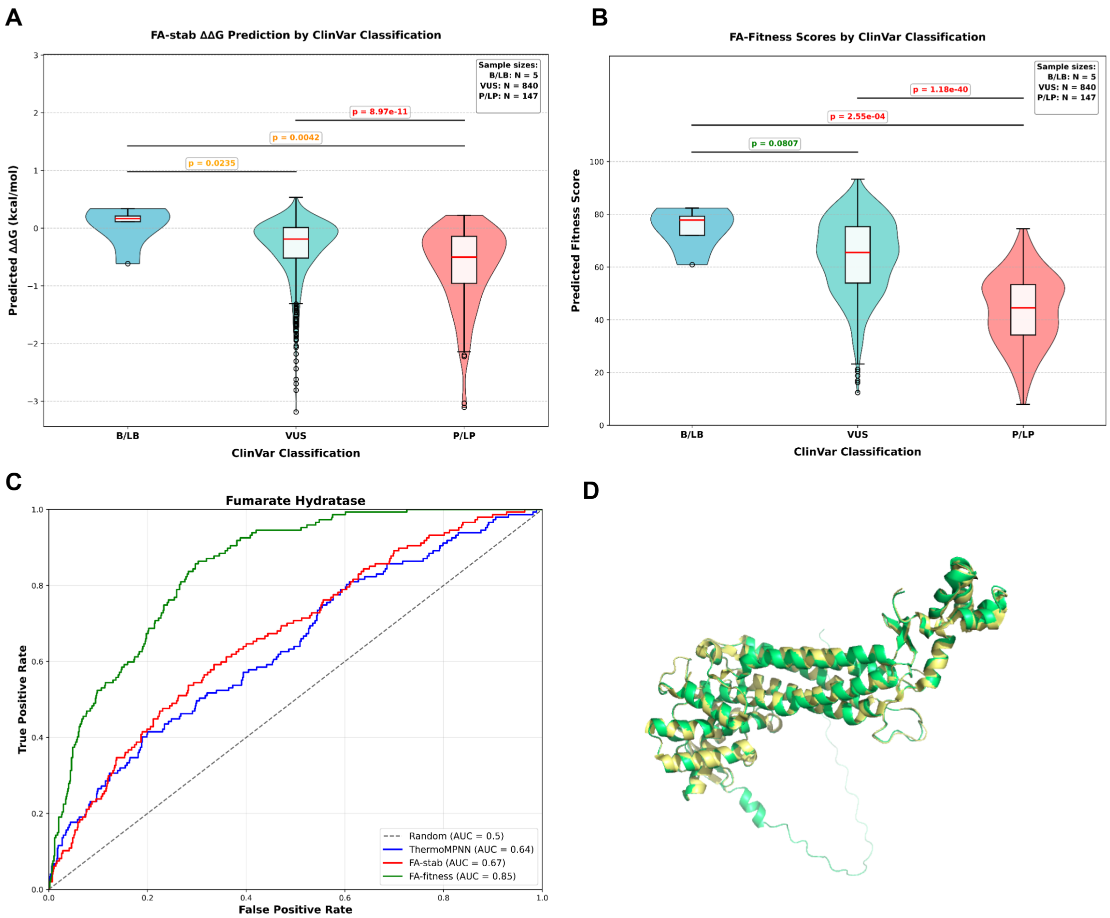


**Figure 4. Distribution of FA-fitness predictions on fumarate hydratase mutations and FA-fitness's potential for disease classification.** (A) Box plot of FA stab $\Delta\Delta G$ predictions for mutations with different ClinVar classifications in fumarate hydratase. (B) Box plot of FA fitness predictions for mutations with different ClinVar classifications in fumarate hydratase. (C) ROC AUC curves of FA fitness, FA stab and ThermoMPNN predictions (Min-Max normalized) for detecting pathogenic mutations (P/LP). P/LP mutations are treated as positive samples, while B/LB and VUS mutations are considered negative samples. (D) The AlphaFold3-predicted structure (green) and the experimentally resolved structure from the PDB (yellow, PDB ID: 5D6B) of fumarate hydratase.

To further assess the performance of FA-fitness in mutation pathogenicity classification, we benchmarked FA-fitness on the ProteinGym 'Clinical Substitutions' dataset, which was originally derived from ClinVar[48] and provides binary labels (pathogenic vs. non-pathogenic) for each variant. After removing proteins with fewer than 20 mutations and those larger than 1000 residues, the final test set contained 498 proteins. FA-fitness approaches a mean AUC of 0.91 across all proteins. As presented in Supplementary Table S6, FA-fitness is the only fitness-based predictor and exceeds almost all zero-shot clinical predictors. MutPred[49] (AUC=0.91) employs disease-associated mutations as its training set, whereas PrimateAI[50] (AUC=0.85) is trained on large-scale primate mutation data. In contrast, models such as Provean[51] (AUC=0.90) and SIFT4G[52] (AUC=0.89) assess mutation deleteriousness by scoring conservation information derived from homologous sequence searches. FA-fitness, however, is not trained on human population mutation data and primarily relies on fitness prediction values for pathogenicity classification of mutations.This result highlights the promise of fitness-based modeling and DMS data for pathogenicity prediction. On the other hand, FA-fitness falls below MSA-based models such as PoET[53] (0.93), TranceptEVE_L (0.93), GEMME (0.93), and EVE[54] (0.92), underscoring the continued importance of evolutionary information in pathogenic variant assessment.

Here, we use fumarate hydratase (Uniprot ID: P07954) as an example to demonstrate the classification capability of FA-fitness in predicting mutation pathogenicity. Fumarate hydratase is an enzyme primarily localized within mitochondria, catalyzing the hydration of fumarate to L-malate. Mutations in this protein may lead to Fumarase deficiency, characterized by abnormal brain development or early-onset dystonia. We collected 992 ClinVar-annotated mutations in fumarate hydratase, including 5 benign/likely benign (B/LB), 147 pathogenic/likely pathogenic (P/LP), and 840 variants of uncertain significance (VUS). As presented in Figure 4, we plotted the predicted $\Delta\Delta$G from FA-stab and the fitness scores from FA-fitness for these three mutation categories as box plots and performed the Mann–Whitney U test to evaluate whether significant differences exist between distinct mutation classes. The FA-fitness scores were normalized via Min-Max normalization and scaled to a percentage range from 0 to 100. The results show that the $\Delta\Delta$G predicted by FA-stab exhibits a significant difference between P/LP and VUS (P = 8.97e-11) and a moderately significant difference between P/LP and B/LB (P = 4.2e-3). In contrast, the fitness scores predicted by FA-fitness show stronger statistical significance between P/LP vs. VUS (P = 1.18e-40) and P/LP vs. B/LB (P = 2.55e-4), indicating that FA-fitness possesses a stronger discriminative ability among ClinVar-defined mutation categories. Furthermore, to evaluate the model's performance in detecting P/LP mutations, we treated P/LP as positive samples and B/LB together with VUS as negative samples, then plotted ROC curves using Min–Max normalized $\Delta\Delta$G values from Fa-fitness, FA-stab and ThermoMPNN. The AUC values of FA-stab and ThermoMPNN were comparable, at 0.67 and 0.64, respectively. In comparison, FA-fitness achieved a notably higher AUC of 0.85, further demonstrating that fitness-based prediction models hold an advantage over stability-based models in predicting mutation pathogenicity.

## Discussion

Backbone-based protein design models, such as ProteinMPNN and ESM-IF, have achieved notable success in protein design. In the field of mutational effect prediction, a number of studies have drawn inspiration from protein inverse-folding models by extracting protein backbone structural information to predict changes in protein stability upon mutation. However, these studies generally overlook the structural changes in protein side chains before and after mutation. In this work, we introduce an all-atom equivariant network to encode both mutant and wild-type protein structures, and construct the protein stability prediction model FA-stab. This model achieves higher correlation coefficients (SCC and PCC) on the S669 test set than other models, including those that use backbone atomic coordinates as structural input. It also surpasses ThermoMPNN on the Myoglobin and Alpha-1-antitrypsin test sets. On the large-scale multi-mutation test set of the orphan protein SeviL, FA-stab outperforms both the backbone-based SPIRED-Stab and the energy-function-based DDGun3D, demonstrating the importance of fully utilizing structural information for stability prediction in proteins lacking homologous sequences.

Furthermore, we extended the all-atom-aware model architecture of FA-stab to the fitness prediction task, training on DMS data and the MegaScale $\Delta\Delta$G Data to obtain the FA-fitness model. Compared to other zero-shot predictors on the ProteinGym DMS Substitutions benchmark, FA-fitness demonstrates strong generalization performance. Moreover, FA-fitness outperforms other clinical predictors on the ProteinGym Clinical Substitutions test set in terms of AUC. In predicting mutation pathogenicity for fumarate hydratase, FA-fitness also surpasses both FA-stab and ThermoMPNN. These results highlight the potential of fitness-based models in mutation pathogenicity assessment. It also suggests that accurate pathogenicity prediction requires the integration of multiple dimensions of information, including stability, fitness (DMS) scores, and evolutionary signals.

Protein–ligand binding stability or affinity is closely related to the structural recognition of protein side chains. Previous protein stability prediction models utilized only the protein backbone information, thus failing to exploit the atomic relative positional information in the side-chain and ligand-binding regions. This study introduces an all-atom equivariant network into protein stability prediction, paving the way for more accurate prediction of stability in protein–ligand complexes. We plan to extend the all-atom equivariant network presented here to the prediction of protein–protein complex stability and protein–small-molecule binding affinity in future work. Furthermore, in the present study, the structural diversity of proteins in the $\Delta\Delta$G training set is relatively limited. The all-atom structural encoder has not undergone large-scale pre-training on comprehensive protein structure databases such as CATH, PDB, or AlphaFoldDB. Pre-training the all-atom encoder on such databases in the future may further enhance the model's ability to predict protein stability.

## Methods

### Feature processing pipeline for the FA-stab model

The all-atom wild-type protein structure is predicted using AlphaFold3. This predicted wild-type structure is then provided as input to FoldX, which generates the corresponding all-atom mutant structure. From the structures predicted by AlphaFold3 and FoldX, we extract the spatial coordinates, atom types (C, N, O, S), per-atom pLDDT confidence scores, and the residue type to which each atom belongs. During subsequent feature processing in FA-stab, the atom type, pLDDT, and residue type information are combined into a unified atomic feature representation.

On the other hand, sequence-based features are derived from protein language models. The multiple sequence alignment (MSA) for the wild-type protein is obtained through the homology search pipeline of AlphaFold3 and subsequently filtered using hhfilter (-diff 128). The MSA for the mutant is generated by replacing the query sequence in the filtered wild-type MSA with the mutant sequence. Evolutionary and fitness information is then extracted using ESM2-650M (for single-sequence embeddings), ESM-1v (for fitness-related embeddings) and MSA Transformer (for co-evolutionary information derived from the MSA).

### Training details for FA-stab and FA-fitness

FA-stab was trained using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 128. FA-stab used L1 loss function, which computes the mean absolute error (MAE) between the predicted and experimental $\Delta\Delta\mathrm{G}$ values. As shown in Equations1–3, $\Delta\Delta\hat{\mathrm{G}}_{\mathrm{dir}}$ and $\Delta\Delta\hat{\mathrm{G}}_{\mathrm{inv}}$ are the $\Delta\Delta\mathrm{G}$ values predicted by FA-stab for the direct and inverse mutations, respectively, while $\Delta\Delta\mathrm{G}_{\mathrm{dir}}$ and $\Delta\Delta\mathrm{G}_{\mathrm{inv}}$ denote the corresponding forward and reverse $\Delta\Delta\mathrm{G}$ labels. $\mathrm{Loss}_{\mathrm{dir}}$ represents the MAE loss of the direct $\Delta\Delta\mathrm{G}$ . The final loss function of FA-stab, $\mathrm{Loss}_{\mathrm{all}}$, is defined as the average of the direct and inverse losses.

$$\mathrm{Loss}_{\mathrm{dir}} = |\Delta\Delta\mathrm{G}_{\mathrm{dir}} - \Delta\Delta\hat{\mathrm{G}}_{\mathrm{dir}}| \tag{1}$$

$$\mathrm{Loss}_{\mathrm{inv}} = |\Delta\Delta\mathrm{G}_{\mathrm{inv}} - \Delta\Delta\hat{\mathrm{G}}_{\mathrm{inv}}| \tag{2}$$

$$\mathrm{Loss}_{\mathrm{all}} = \frac{(\mathrm{Loss}_{\mathrm{dir}} + \mathrm{Loss}_{\mathrm{inv}})}{2} \tag{3}$$

FA-fitness was trained with an initial learning rate of 0.001 using the Adam optimizer. The ReduceLROnPlateau scheduler (patience = 3, factor = 0.5) was employed to adjust the learning rate during training. The model utilized a Soft Spearman Loss[12,55] to penalize discrepancies in the rank order between the predicted fitness scores and the true labels for all single mutations within each protein. Training of both FA-stab and FA-fitness was conducted with on NVIDIA A100 80GB GPUs.

#### Training set for FA-stab

**Mega-scale $\Delta\Delta$G Dataset** (cDNA Proteolysis Dataset)[56] is a high-throughput dataset created for measuring mutation-induced changes in protein thermal stability. This method involved constructing a library where mutant proteins were covalently linked to corresponding cDNA fragments. Less stable mutants were preferentially digested by proteolysis, and sequencing the surviving cDNA tags yielded proteolysis rate data, from which $\Delta G$ values were derived via Bayesian inference. Using MMseqs2[57], we removed proteins with >30% sequence similarity to those in our test sets. The final training set for FA-stab comprised approximately 415K single and double mutations ($\Delta\Delta$G data) across 334 proteins.

#### Training set for FA-fitness

The training set for FA-fitness comprises two components. The first part consists of stability data from the aforementioned Mega-scale $\Delta\Delta$G Dataset, which includes 334 proteins. The second part incorporates fitness data for 62 proteins sourced from the MaveDB[41] and DeepSequence dataset[42]. Only single-point mutations were selected from these sources for training FA-fitness.

#### Test sets for protein stability prediction

**S669**[17] is one of the most widely adopted test sets for $\Delta\Delta$G prediction. It comprises 669 manually curated single-point mutations collected from the ThermoMutDB[58] database.

**Myoglobin** dataset[38] serves as a benchmark for predicting the stability effects of mutations in the oxygen-binding protein. After removing duplicate entries from the original set of 134 single mutations, 113 unique mutations were retained for analysis.

**p53** test set comprises $\Delta\Delta$G data for 46 single-point mutations within the core domain of cellular tumor antigen p53 (Uniprot: P04637), sourced from the FireProt database[59].

**Alpha-1-antitrypsin** test set contains 43 single mutations in the human serine protease inhibitor Alpha-1-antitrypsin (Uniprot: P01009), where $\Delta\Delta$G data of these mutations were obtained from the FireProt database[59].

**'Orphan' Protein SeviL** (PDB ID: 6LF2) is a lectin that binds saccharides of ganglioside GM1b, which makes it a promising candidate for detecting immune and cancer cells displaying asialo-GM1[60]. SeviL adopts a $\beta$-trefoil fold but exhibits very weak sequence similarity with other proteins. The Sibs-Seq experiment generated a large-scale fitness score dataset, reflecting mutational stability for SeviL variants[35]. This dataset comprises 891 single mutations, 20,332 double mutations, and 37,911 triple and higher-order mutations. In this study, we used the fitness data of these mutants as a test set to evaluate the predictive capability of models for stability changes induced by multiple mutations.

All stability test set proteins have less than 30% sequence similarity with training proteins.

### Test set for FA-fitness

**ProteinGym Fitness test set**. This dataset was derived from the 'DMS Substitutions' section of ProteinGym[61]. We performed sequence similarity checks between proteins in ProteinGym and those in the aforementioned fitness training set using MMseqs2, retaining only those proteins from ProteinGym with less than 30% sequence similarity to the training proteins. This resulted in 71 assays, which were used as a zero-shot test set for FA-fitness.

**ProteinGym ClinVar test set**. ProteinGym organized mutation data from ClinVar[48] into its 'Clinical Substitutions' section, where each mutation was assigned a binary label, pathogenic (1) or non-pathogenic (0). We filtered out proteins with fewer than 20 mutations and those with sequence lengths exceeding 1000 amino acids from the original ProteinGym 'Clinical Substitutions' dataset, resulting in a final set of 498 proteins. This dataset was used to evaluate the classification performance of FA-fitness in predicting mutation pathogenicity.

### Evaluation metrics

**Root Mean Square Error** (RMSE) is a standard metric for quantifying the average magnitude of prediction errors. It is calculated as the square root of the average squared differences between predicted values ($\hat{y}$) and corresponding label values ($y$), as shown in Equation 4. Due to the squaring step, RMSE penalizes larger errors more heavily than smaller ones, making it particularly sensitive to outliers.

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{4}$$

**Mean Absolute Error** (MAE) measures the average absolute difference between predicted and ground-truth values. Unlike RMSE, it does not square the errors, treating all deviations linearly and equally. Consequently, MAE is less sensitive to extreme outliers.

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{5}$$

**Pearson correlation coefficient** (PCC) quantifies the strength and direction of a linear relationship between two variables. It is calculated as the ratio of the covariance of the two variables to the product of their standard deviations, as shown in Equation 6. The PCC values of 1, 0, and -1 correspond to a perfect positive linear correlation, no linear association, and a perfect negative linear correlation, respectively.

$$\text{PCC} = \frac{Cov(y, \hat{y})}{\sigma_y \sigma_{\hat{y}}}, \tag{6}$$

where $Cov$ stands for the covariance between model-predicted value y and label $\hat{y}$. The $\sigma$ denotes the standard deviation.

**Spearman correlation coefficient** (SCC) measures the strength and direction of a monotonic relationship between two variables. Unlike Pearson correlation coefficient, SCC is computed using the rank-ordered values of each variable

($Rank(y)$ and $Rank(\hat{y})$) as shown in the Equation 7. This rank-based approach makes SCC less sensitive to outliers and more appropriate for assessing ordinal associations or non-linear trends in data.

$$\text{SCC} = \frac{Cov(Rank(y), Rank(\hat{y}))}{\sigma_{Rank(y)}\sigma_{Rank(\hat{y})}} \tag{7}$$

**Kendall correlation coefficient** (KCC) is a nonparametric measure of association based on the ranks of two variables. It quantifies the probability that pairs of observations are ordered consistently (concordant) versus inconsistently (discordant) across the two rankings (see Equation8).

$$\text{KCC} = \frac{n_c - n_d}{n_c + n_d}, \tag{8}$$

where $n_c$ and $n_d$ represent the number of concordant and discordant pairs, respectively.

**Area under the receiver operating characteristic curve** (AUC) is a performance metric for binary classifiers. The value of AUC, bounded between 0.5 (random guessing) and 1.0 (perfect separation), summarizes model performance across all thresholds, offering a robust, threshold-independent measure of predictive accuracy.

## Code availability

The FA-stab and FA-fitness are implemented in PyTorch and e3nn-torch[13]. All codes including the full training scripts are freely downloadable at

## Author contribution

Y. Chen, J. Zhan, and Y. Zhou designed the study. Y. Chen implemented the model codes, conducted model training and performance testing. Y. Chen performed data analysis and curation. Y. Chen, J. Zhan, and Y. Zhou contributed to writing the manuscript. J. Zhan and Y. Zhou supervised this project. Y. Zhou provided the funding support.


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## Ethics declarations

### Competing interests

The authors declare no competing interests.

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# Supplementary Information

# An all-atom structure aware method for versatile prediction of protein stability changes and broad mutational effects

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# 1 The Model Architecture of the FA-fitness


**Figure S1. chematic of the FA-fitness model architecture.** The wild-type protein structure predicted by AlphaFold3 is processed through an all-atom equivariant network to extract atom-level spatial information, which are then aggregated into residue-level structural features. These features serve as node inputs to a Graph Attention Network (GAT), while the contact probability and predicted aligned error (pAE) from AlphaFold3 provide edge information. The GAT refines these representations and outputs a final structural embedding. This structural embedding is subsequently integrated with wild-type sequence features derived from a protein language model to generate the predicted fitness logits.

## 2 $\Delta\Delta\mathrm{G}$ prediction performace on inverse mutations within each protein of the S669 test set

**Table S1. Evaluation of $\Delta\Delta\mathrm{G}$ prediction of inverse mutations within each protein on the S669 dataset**

| **Method** | **SCC** ↑ | **PCC** ↑ | **KCC** ↑ | **RMSE** ↓ | **MAE** ↓ |
|---|---|---|---|---|---|
| *Sequence-based* | | | | | |
| **SAAFEC-SEQ**[1] | -0.16 ± 0.34 | -0.14 ± 0.32 | -0.10 ± 0.22 | 2.06 ± 0.78 | 1.81 ± 0.77 |
| **MUPro**[2] | 0.10 ± 0.25 | 0.09 ± 0.26 | 0.07 ± 0.18 | 2.03 ± 0.63 | 1.78 ± 0.61 |
| **I-Mutant3.0-Seq**[3] | 0.12 ± 0.20 | 0.10 ± 0.22 | 0.09 ± 0.13 | 1.85 ± 0.59 | 1.60 ± 0.56 |
| **INPS-Seq**[4] | 0.38 ± 0.27 | 0.43 ± 0.25 | 0.27 ± 0.21 | 1.15 ± 0.50 | 0.92 ± 0.44 |
| **ACDC-NN-Seq**[5] | 0.39 ± 0.28 | 0.44 ± 0.28 | 0.29 ± 0.21 | 1.14 ± 0.47 | 0.90 ± 0.43 |
| **DDGun**[6] | 0.46 ± 0.24 | 0.47 ± 0.26 | 0.35 ± 0.18 | 1.40 ± 0.68 | 1.12 ± 0.60 |
| **PROSTATA**[7] | 0.50 ± 0.26 | 0.53 ± 0.26 | 0.36 ± 0.20 | 1.06 ± 0.49 | 0.83 ± 0.43 |
| *Structure-based* | | | | | |
| **I-Mutant3.0**[3] | -0.01 ± 0.25 | -0.03 ± 0.26 | -0.01 ± 0.17 | 1.98 ± 0.59 | 1.71 ± 0.57 |
| **mCSM**[8] | 0.03 ± 0.24 | 0.04 ± 0.26 | 0.03 ± 0.17 | 1.97 ± 0.52 | 1.69 ± 0.50 |
| **SDM**[9] | 0.05 ± 0.27 | 0.05 ± 0.29 | 0.03 ± 0.21 | 1.78 ± 0.56 | 1.42 ± 0.48 |
| **DUET**[10] | 0.04 ± 0.26 | 0.06 ± 0.29 | 0.03 ± 0.19 | 1.79 ± 0.49 | 1.49 ± 0.45 |
| **PoPMuSiC**[11] | 0.07 ± 0.28 | 0.05 ± 0.28 | 0.05 ± 0.20 | 1.76 ± 0.46 | 1.47 ± 0.47 |
| **MAESTRO**[12] | 0.15 ± 0.27 | 0.13 ± 0.28 | 0.12 ± 0.19 | 1.75 ± 0.51 | 1.45 ± 0.50 |
| **ThermoNet**[13] | 0.24 ± 0.30 | 0.33 ± 0.28 | 0.16 ± 0.22 | 1.28 ± 0.40 | 1.04 ± 0.38 |
| **Dynamut**[14] | 0.27 ± 0.28 | 0.27 ± 0.26 | 0.18 ± 0.19 | 1.34 ± 0.44 | 1.06 ± 0.41 |
| **FoldX**[15] | 0.28 ± 0.30 | 0.24 ± 0.33 | 0.20 ± 0.22 | 1.75 ± 0.80 | 1.28 ± 0.59 |
| **INPS3D**[4] | 0.37 ± 0.23 | 0.37 ± 0.25 | 0.26 ± 0.17 | 1.43 ± 0.42 | 1.15 ± 0.39 |
| **ACDC-NN**[5] | 0.38 ± 0.30 | 0.42 ± 0.30 | 0.28 ± 0.22 | 1.12 ± 0.46 | 0.88 ± 0.43 |
| **PremPS**[16] | 0.39 ± 0.29 | 0.39 ± 0.32 | 0.29 ± 0.21 | 1.16 ± 0.53 | 0.94 ± 0.48 |
| **DDGun3D**[6] | 0.42 ± 0.29 | 0.43 ± 0.30 | 0.32 ± 0.23 | 1.25 ± 0.57 | 0.97 ± 0.50 |
| **DDMut**[17] | 0.45 ± 0.35 | 0.49 ± 0.30 | 0.33 ± 0.28 | 1.12 ± 0.49 | 0.92 ± 0.43 |
| *Structure-sequence-hybrid* | | | | | |
| **GeoDDG-3D**[18] | 0.50 ± 0.27 | 0.51 ± 0.26 | 0.38 ± 0.21 | 1.11 ± 0.41 | 0.87 ± 0.39 |
| **GeoDDG-Seq**[18] | 0.51 ± 0.27 | 0.52 ± 0.28 | 0.39 ± 0.21 | 1.08 ± 0.42 | 0.85 ± 0.37 |
| **GeoStab v2**[19] | 0.55 ± 0.39 | 0.54 ± 0.40 | 0.43 ± 0.31 | 1.06 ± 0.51 | 0.84 ± 0.46 |
| **SPIRED-Stab**[19] | 0.58 ± 0.33 | 0.55 ± 0.34 | **0.46** ± 0.26 | 1.07 ± 0.51 | **0.82** ± 0.45 |
| **FA-stab** | **0.59** ± 0.18 | **0.58** ± 0.14 | 0.43 ± 0.14 | **1.04** ± 0.45 | 0.87 ± 0.41 |

All metrics are evaluated on a per-protein basis, with the final result presented as the mean ± standard deviation across all proteins. For each metric, the top-performing model is identified by comparing these mean values and is indicated in bold.

## 3 Distribution of predicted $\Delta\Delta G$ and Sibs-Seq fitness scores of SeviL mutations

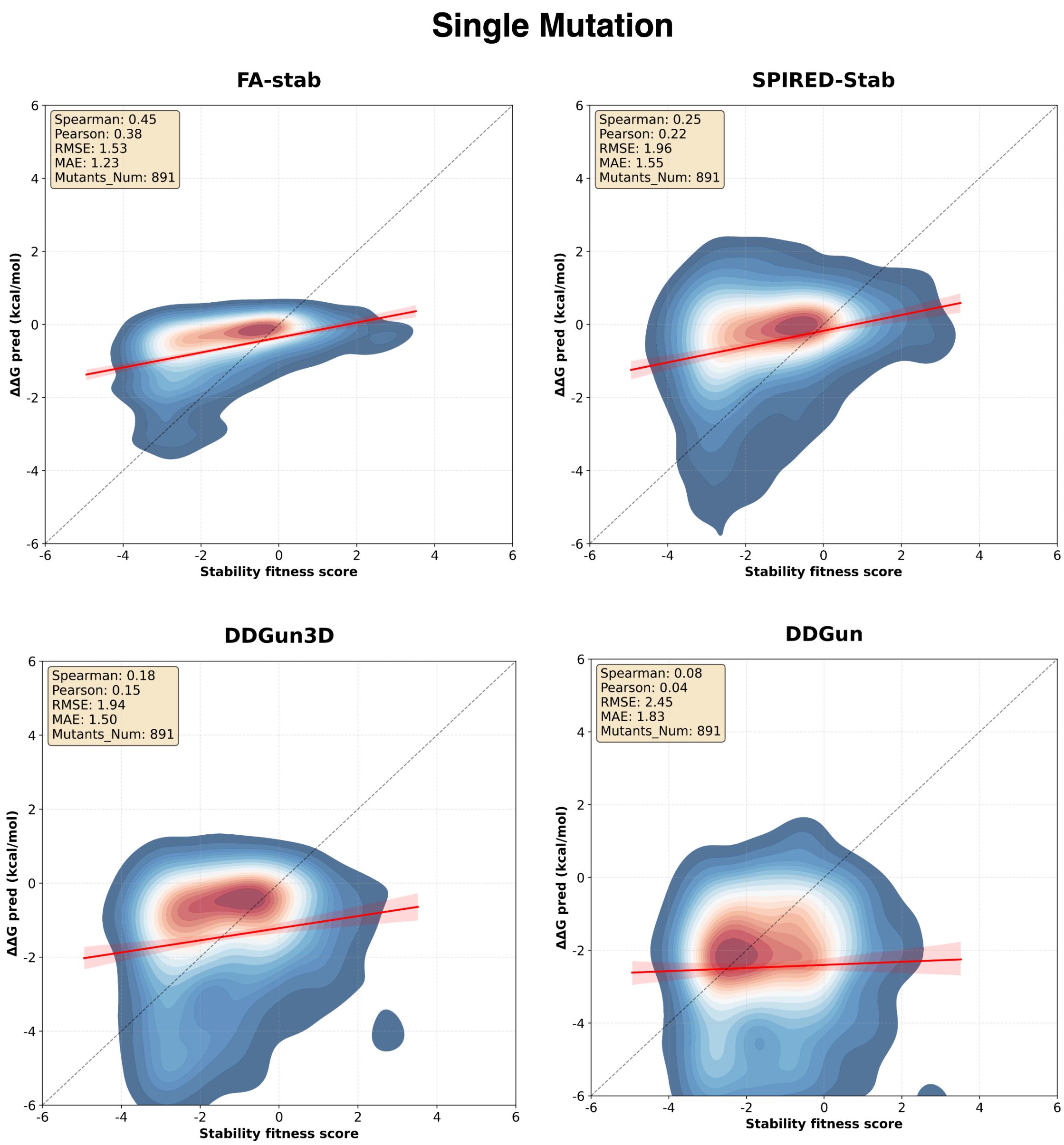


**Figure S2. Density plots of predicted $\Delta\Delta G$ versus experimental fitness scores for single mutations in SeviL.** The horizontal axis represents the fitness score from Sibs-Seq experiments, reflecting the stability of each single mutation. The vertical axis shows the predicted $\Delta\Delta G$ (kcal/mol) from each model. The figure presents results from four models: FA-stab, SPIRED-Stab, DDGun3D and DDGun. Each sub-plot legend reports the corresponding SCC, PCC, RMSE and MAE. The total number of single mutations is 891.

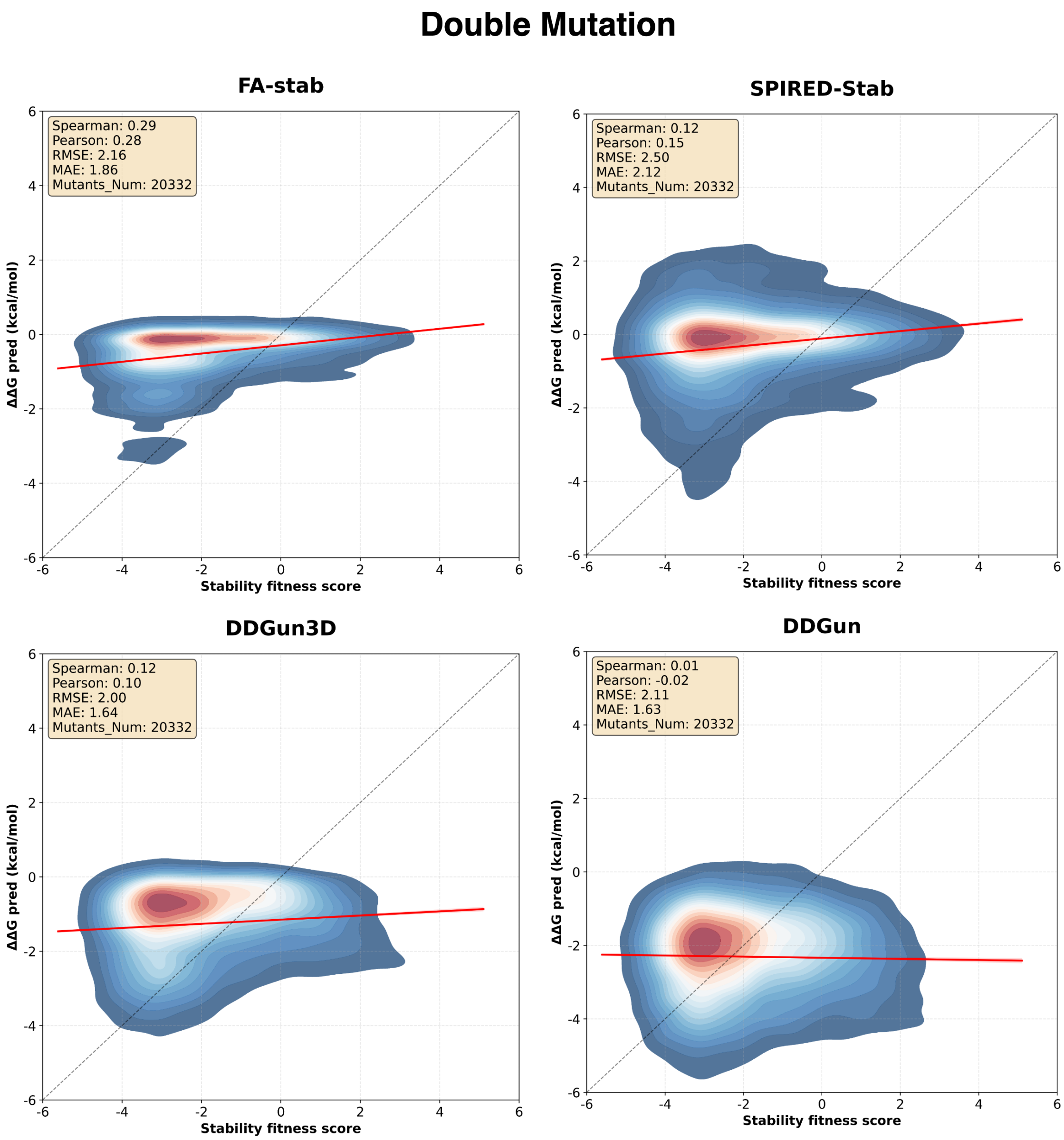


**Figure S3. Density plots of predicted $\Delta\Delta$G versus experimental fitness scores for double mutations in SeviL.** The horizontal axis represents the fitness score from Sibs-Seq experiments, reflecting the stability of each double mutation. The vertical axis shows the predicted $\Delta\Delta$G (kcal/mol) from each model. The figure presents results from four models: FA-stab, SPIRED-Stab, DDGun3D and DDGun. Each sub-plot legend reports the corresponding SCC, PCC, RMSE and MAE. The total number of double mutations is 20332.

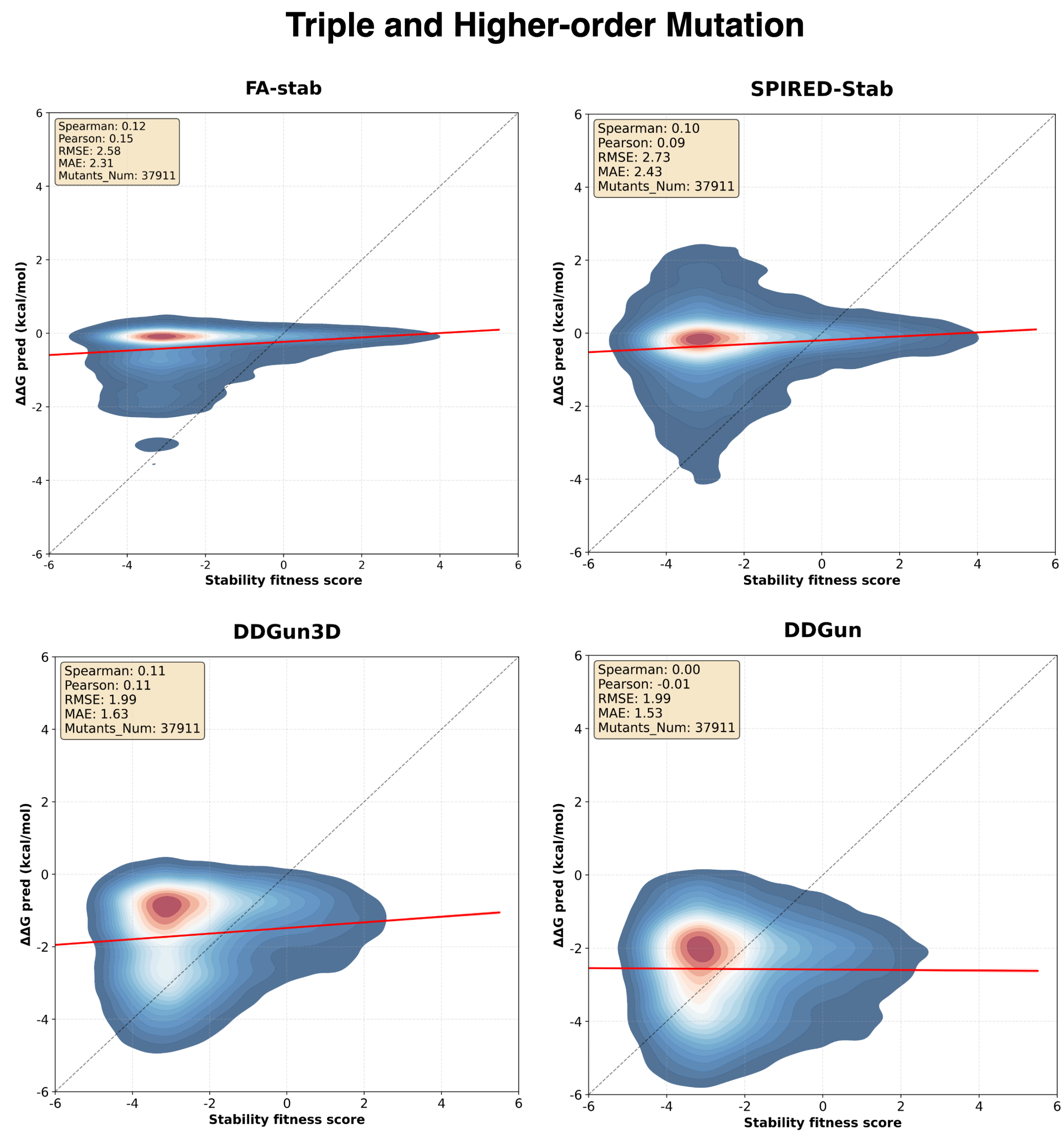


**Figure S4. Density plots of predicted $\Delta\Delta$G versus experimental fitness scores for triple and higer-order mutations in SeviL.** The horizontal axis represents the fitness score from Sibs-Seq experiments, reflecting the stability of each mutation. The vertical axis shows the predicted $\Delta\Delta$G (kcal/mol) from each model. The figure presents results from four models: FA-stab, SPIRED-Stab, DDGun3D and DDGun. Each sub-plot legend reports the corresponding SCC, PCC, RMSE and MAE. The total number of triple and higer-order mutations is 37911.

# 4 Performace on mutations of all the proteins in Sibs-Seq dataset

**Table S2. Spearman correlation coefficients for single mutations in Sibs-Seq dataset**

| Protein | FA-stab | SPIRED-Stab | DDGun3D | DDGun | Mutation Number |
|---|---|---|---|---|---|
| 6LF2-102 | 0.45 | 0.25 | 0.18 | 0.08 | 891 |
| 6LYD-252 | 0.37 | 0.08 | 0.27 | 0.29 | 813 |
| 6TN7-252 | 0.45 | 0.48 | 0.31 | 0.34 | 720 |
| 7CHQ-252 | 0.48 | 0.20 | 0.33 | 0.22 | 872 |
| 7F0K-103 | 0.25 | 0.24 | 0.23 | 0.18 | 632 |
| 7ZU3-102 | -0.04 | 0.02 | -0.01 | -0.03 | 745 |
| 8EB9-252 | 0.24 | 0.06 | 0.19 | 0.23 | 544 |
| Tue1-252 | 0.22 | 0.04 | 0.11 | 0.17 | 1042 |
| Mean | 0.30 | 0.17 | 0.20 | 0.19 | 782 |

**Table S3. Spearman correlation coefficients for double mutations in Sibs-Seq dataset**

| Protein | FA-stab | SPIRED-Stab | DDGun3D | DDGun | Mutation Number |
|---|---|---|---|---|---|
| 6LF2-102 | 0.29 | 0.12 | 0.12 | 0.01 | 20332 |
| 6LYD-252 | 0.19 | 0.03 | 0.16 | 0.17 | 30291 |
| 6TN7-252 | 0.36 | 0.37 | 0.38 | 0.37 | 11288 |
| 7CHQ-252 | 0.37 | 0.23 | 0.34 | 0.20 | 53816 |
| 7F0K-103 | 0.17 | 0.20 | 0.19 | 0.13 | 15078 |
| 7ZU3-102 | -0.35 | -0.26 | -0.19 | -0.26 | 4769 |
| 8EB9-252 | 0.09 | 0.00 | 0.12 | 0.11 | 1022 |
| Tue1-252 | 0.09 | -0.03 | -0.02 | 0.04 | 19046 |
| Mean | 0.15 | 0.08 | 0.14 | 0.10 | 19455 |

**Table S4. Spearman correlation coefficients for triple and higher-order multiple mutations in Sibs-Seq dataset**

| Protein | FA-stab | SPIRED-Stab | DDGun3D | DDGun | Mutation Number |
|---|---|---|---|---|---|
| 6LF2-102 | 0.18 | 0.10 | 0.11 | 0.00 | 37911 |
| 6LYD-252 | 0.03 | -0.00 | 0.01 | 0.03 | 1163 |
| 6TN7-252 | 0.43 | 0.06 | 0.32 | 0.30 | 942 |
| 7CHQ-252 | 0.29 | 0.09 | 0.30 | 0.14 | 192443 |
| 7F0K-103 | 0.08 | 0.14 | 0.21 | 0.16 | 37026 |
| 7ZU3-102 | -0.10 | -0.09 | -0.05 | -0.12 | 22583 |
| 8EB9-252 | -0.07 | -0.11 | -0.04 | -0.06 | 134 |
| Tue1-252 | 0.12 | -0.00 | 0.01 | 0.09 | 15492 |
| Mean | 0.12 | 0.02 | 0.11 | 0.07 | 38462 |

## 5 Evaluation of FA-fitness on ProeinGym test test

**Table S5. Evaluation on ProteinGym DMS Substitutions section (zero-shot)**

| Model | Mean | std | Model Type |
|---|---|---|---|
| VenusREM | 0.49 | 0.13 | Structure & MSA |
| S3F_MSA | 0.49 | 0.13 | Structure & MSA |
| S2F_MSA | 0.48 | 0.13 | Structure & MSA |
| ProSST-2048 | 0.48 | 0.14 | Single sequence & Structure |
| **FA-fitness** | **0.48** | **0.16** | **Structure & MSA** |
| ESCOTT | 0.47 | 0.13 | Structure & MSA |
| PoET | 0.47 | 0.13 | MSA |
| ProSST-4096 | 0.46 | 0.16 | Single sequence & Structure |
| TranceptEVE_L | 0.46 | 0.14 | MSA |
| VespaG | 0.46 | 0.14 | Single sequence |
| GEMME | 0.45 | 0.13 | MSA |
| S3F | 0.45 | 0.14 | Single sequence & Structure |
| RSALOR | 0.45 | 0.15 | Structure & MSA |
| EVE_ensemble | 0.44 | 0.14 | MSA |
| S2F | 0.44 | 0.15 | Single sequence & Structure |
| VESPA | 0.44 | 0.14 | Single sequence |
| ProtSSN_ensemble | 0.44 | 0.15 | Single sequence & Structure |
| MSA_Transformer_ensemble | 0.43 | 0.15 | MSA |
| ESM3 | 0.42 | 0.15 | Single sequence, Structure & Function annotations |
| ESM2_3B | 0.42 | 0.17 | Single sequence |
| ESM2_650M | 0.41 | 0.18 | Single sequence |
| ESM2_15B | 0.41 | 0.16 | Single sequence |
| xTrimoPGLM-3B-MLM | 0.40 | 0.17 | Single sequence |
| Progen2_xlarge | 0.79 | 0.15 | Single sequence |
| ESM1v_ensemble | 0.39 | 0.18 | Single sequence |
| ESM1b | 0.37 | 0.18 | Single sequence |
| ESM-IF1 | 0.36 | 0.19 | Structure |
| ProteinMPNN | 0.19 | 0.15 | Structure |

Here, we compared 71 assays from the ProteinGym DMS substitutions dataset whose corresponding proteins exhibit less than 30% sequence similarity to the proteins in the FA-fitness training set.

**Table S6. Evaluation on ProteinGym Clinical Substitutions section (zero-shot)**

| Model | AUC | Model Type |
|---|---|---|
| PoET(200M) | 0.93 | MSA |
| TranceptEVE_L | 0.93 | MSA |
| GEMME | 0.93 | MSA |
| EVE | 0.92 | MSA |
| **FA-fitness** | **0.91** | **Fitness Predictor** |
| MutPred | 0.91 | Clinical Predictor |
| ESM1b | 0.90 | Single Sequence |
| Provean | 0.90 | Clinical Predictor |
| MutationAssessor | 0.89 | Clinical Predictor |
| SIFT4G | 0.89 | Clinical Predictor |
| SIFT | 0.88 | Clinical Predictor |
| PrimateAI | 0.85 | Clinical Predictor |
| LIST-S2 | 0.85 | Clinical Predictor |
| LRT | 0.80 | Clinical Predictor |

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